Simulation-based verification of bfpwr

Samuel Pawel
František Bartoš

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This vignette summarizes the simulation-based verification checks for **bfpwr**. The report focuses on the verification settings and results: the simulated designs, analysis priors, evidence thresholds, sampling schedules, and the size of any package calculation differences relative to the simulation targets.

1 z-Test

Simulated Design Priors

Table 1: Simulated design priors for the z-Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	USD
Z1	Point(0)	10 to 500 by 10	10000	1
Z2	Point(0.2)	10 to 500 by 10	10000	1
Z3	Point(0.5)	10 to 500 by 10	10000	1
Z4	Normal(0.5, 0.1)	10 to 500 by 10	10000	1
Z5	Normal(0.2, 0.3)	10 to 500 by 10	10000	1
Z6	Normal(0.2, 0.5)	10 to 500 by 10	10000	1
Z7	Normal(0, 0.5)	10 to 500 by 10	10000	1
Z8	Point(-0.3)	10 to 500 by 10	10000	1
Z9	Point(0)	10 to 500 by 10	10000	50
Z10	Point(10)	10 to 500 by 10	10000	50
Z11	Normal(10, 20)	10 to 500 by 10	10000	50
Z12	Point(0)	10 to 10000 by 10	10000	1
Z13	Point(0.2)	10 to 10000 by 10	10000	1
Z14	Normal(0.2, 0.1)	10 to 10000 by 10	10000	1
Z15	Normal(0.2, 0.5)	10 to 10000 by 10	10000	1

Analysis Priors

Table 2: Analysis priors used for the z-Test verification run. Designs lists the matching short IDs from the design-prior table.

Function	H1	H0	Designs
bf01	Normal(0, 0.707)	Point(0)	Z1-Z8, Z12-Z15
bf01	Normal(0, 20)	Point(0)	Z9-Z11
bf01	Normal(0.2, 0.3)	Point(0)	Z1-Z8, Z12-Z15

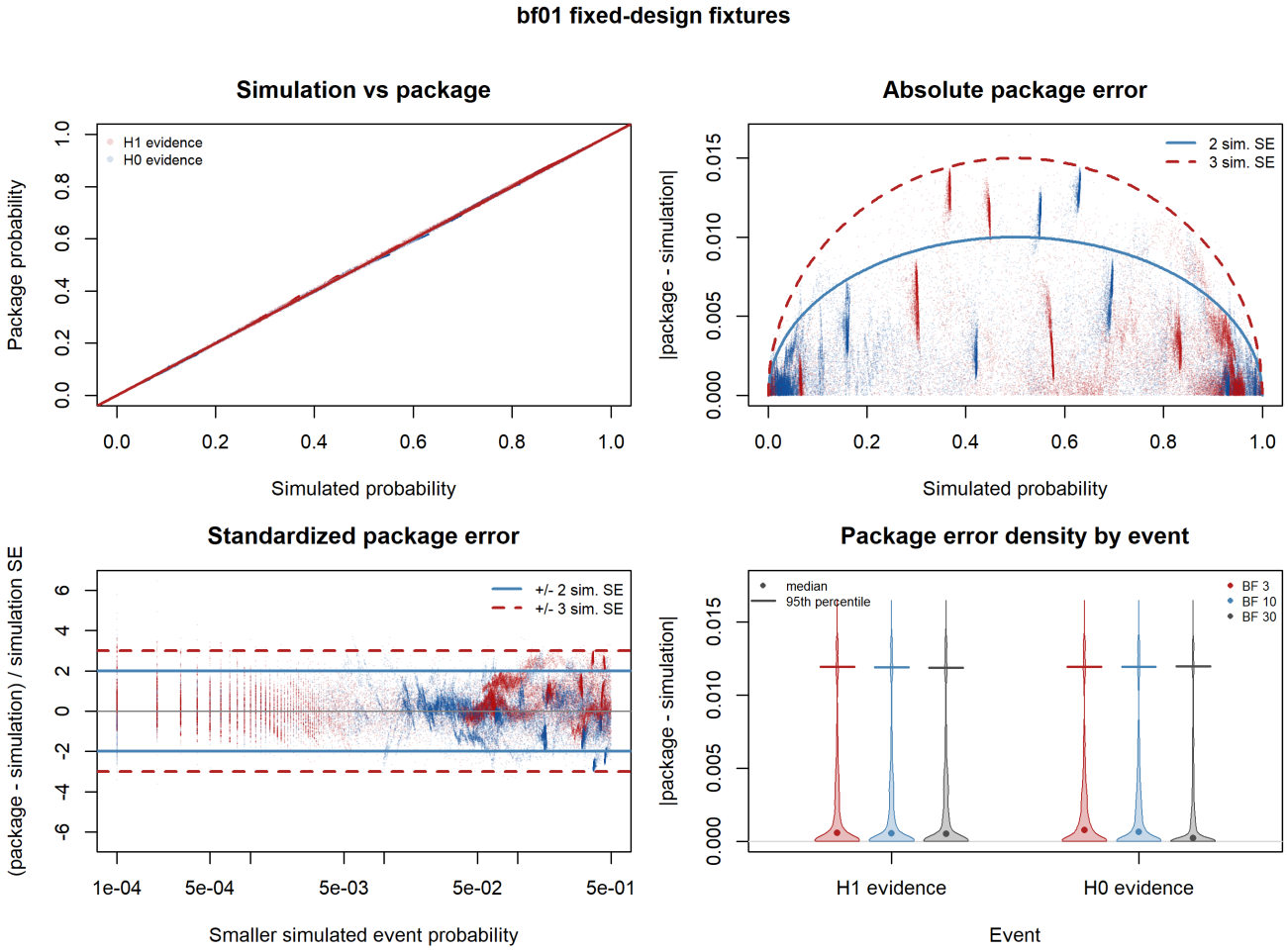
bf01	Normal(0.2, 0.707)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Normal(10, 20)	Point(0)	Z9-Z11
bf01	Point(0.2)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Point(10)	Point(0)	Z9-Z11
dirbf01	$\text{Normal}_+(0, 0.707)$	Point(0)	Z1-Z8, Z12-Z15
dirbf01	$\text{Normal}_+(0, 20)$	Point(0)	Z9-Z11
dirbf01	$\text{Normal}_+(0.2, 0.707)$	Point(0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment(0.354)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(0.354)	Point(0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment(0.707)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(7.071)	Point(0)	Z9-Z11

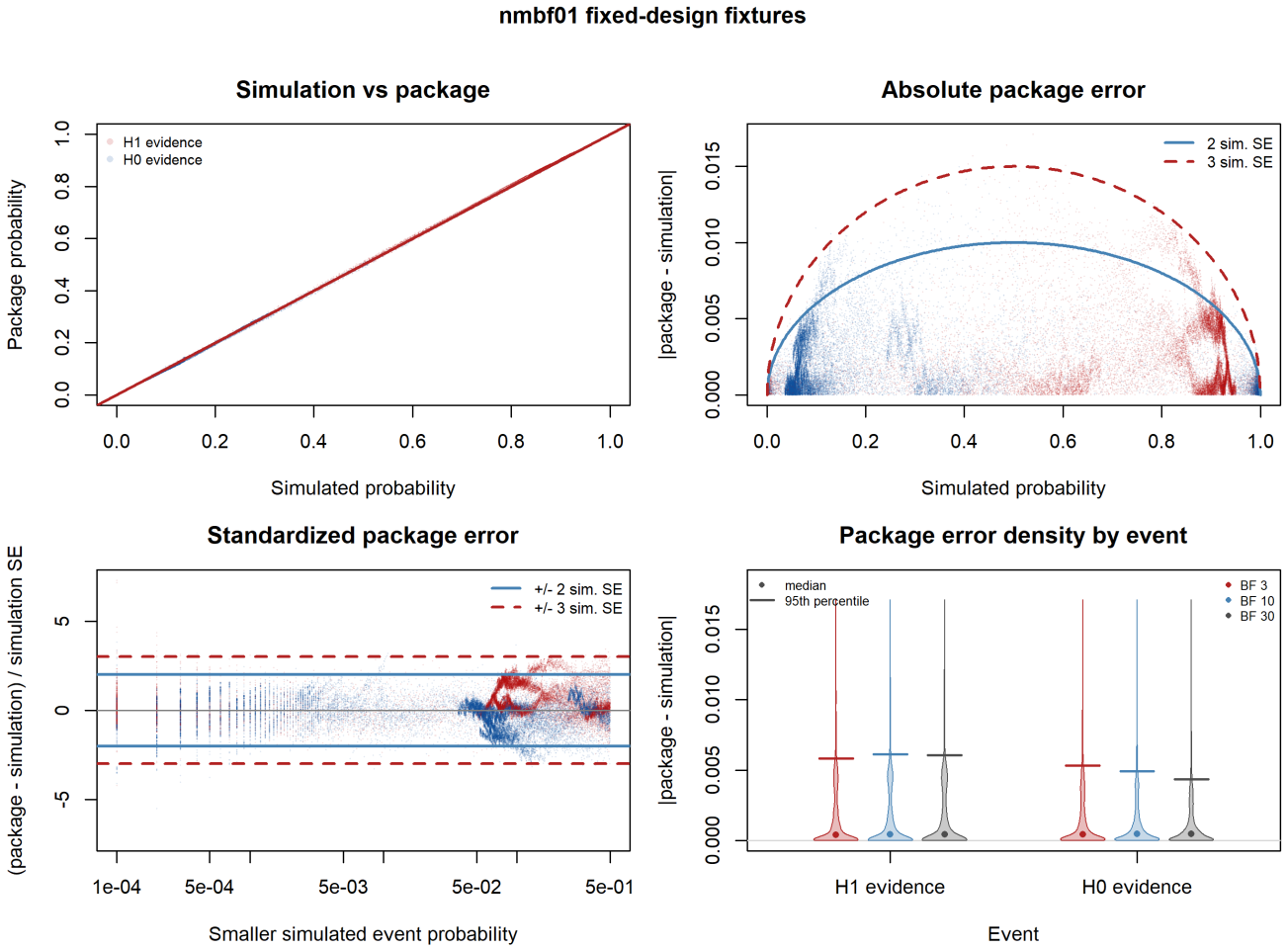
1.1 Fixed-Design

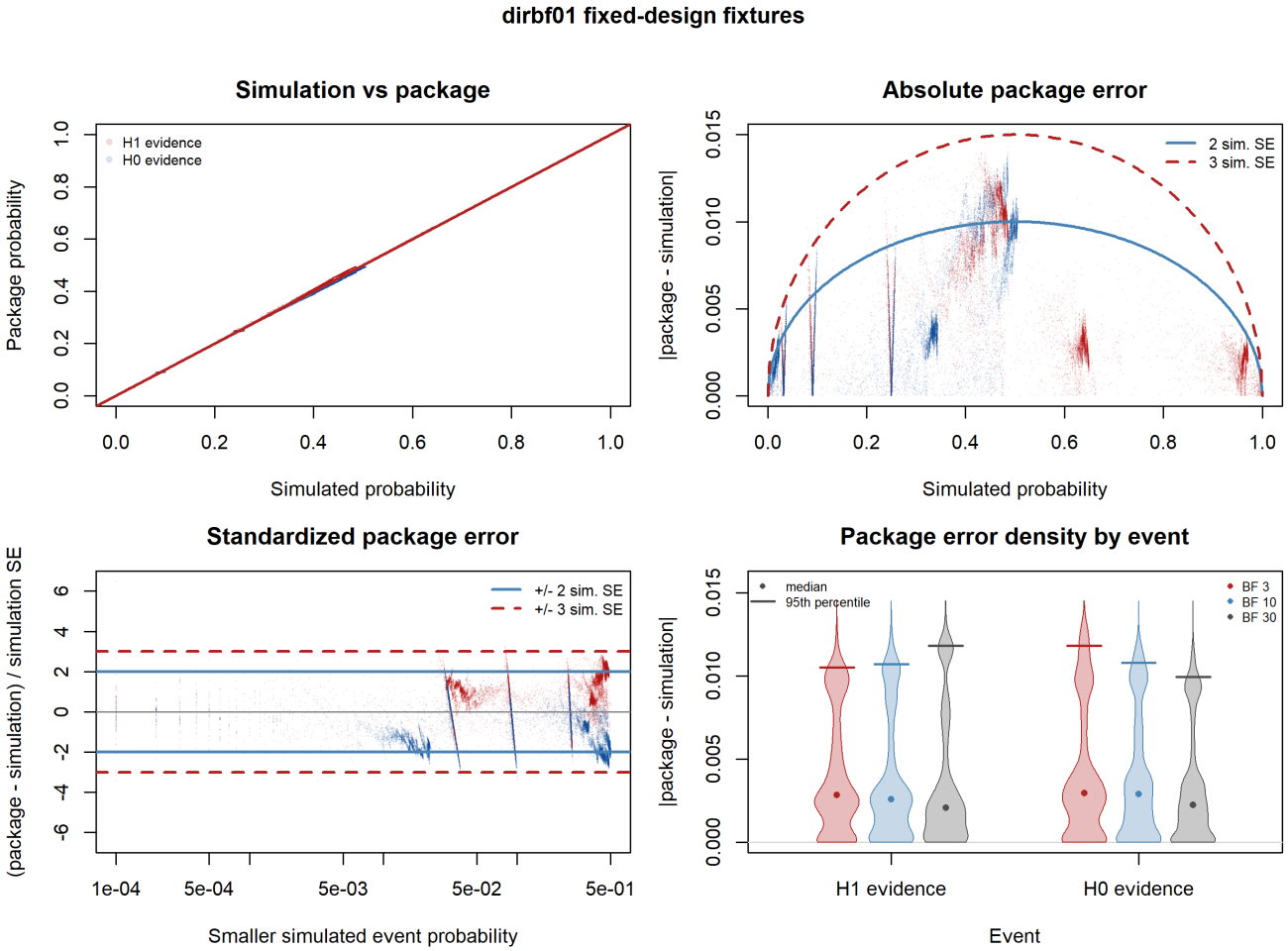
Table 3: Fixed-design fixture setups for z-Test analyses.

Function	Analysis priors	Evidence thresholds
bf01	73	BF 3, 10, 30
nmbf01	35	BF 3, 10, 30
dirbf01	23	BF 3, 10, 30

The fixed-design plots validate package probability calculations against the simulated event frequencies. Cached diagnostic rows are shown; for computationally expensive reference families, the plotted cache can be a stratified subset of the full fixture summary. The panels compare simulated and package probabilities, show absolute and standardized package error, and show package-error densities by event.


bf01 fixed-design fixtures
Simulation vs package
H1 evidence
H0 evidence
Package probability
Simulated probability
Absolute package error
2 sim. SE
3 sim. SE
|package - simulation|
Simulated probability
Standardized package error
+/- 2 sim. SE
+/- 3 sim. SE
(package - simulation) / simulation SE
Smaller simulated event probability
Package error density by event
median
95th percentile
BF 3
BF 10
BF 30
|package - simulation|
H1 evidence
H0 evidence
Event


nmbf01 fixed-design fixtures
Simulation vs package
H1 evidence
H0 evidence
Package probability
Simulated probability
Absolute package error
2 sim. SE
3 sim. SE
|package - simulation|
Simulated probability
Standardized package error
+/- 2 sim. SE
+/- 3 sim. SE
(package - simulation) / simulation SE
Smaller simulated event probability
Package error density by event
median
95th percentile
BF 3
BF 10
BF 30
|package - simulation|
H1 evidence
H0 evidence
Event



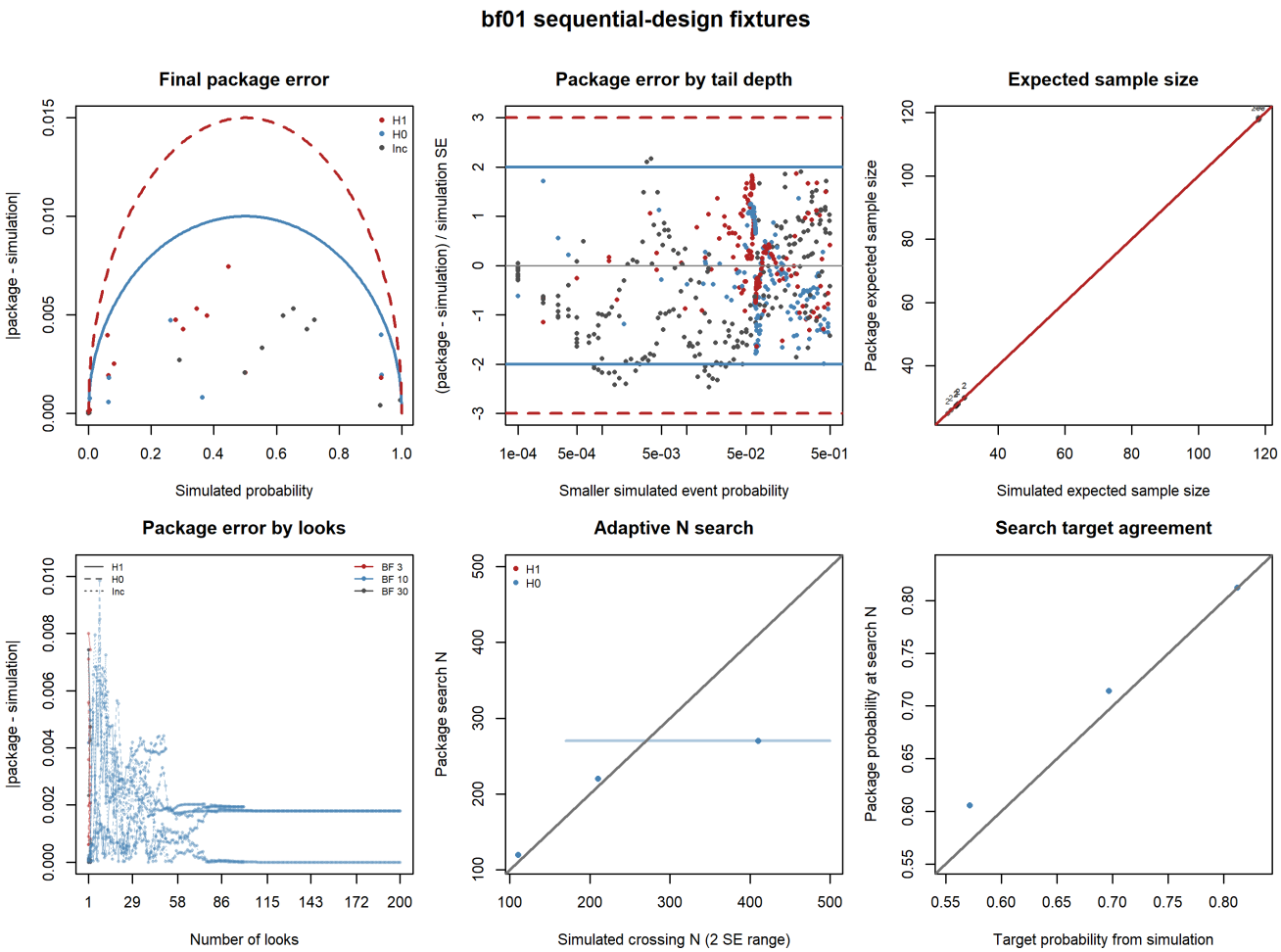
1.2 Sequential-Design

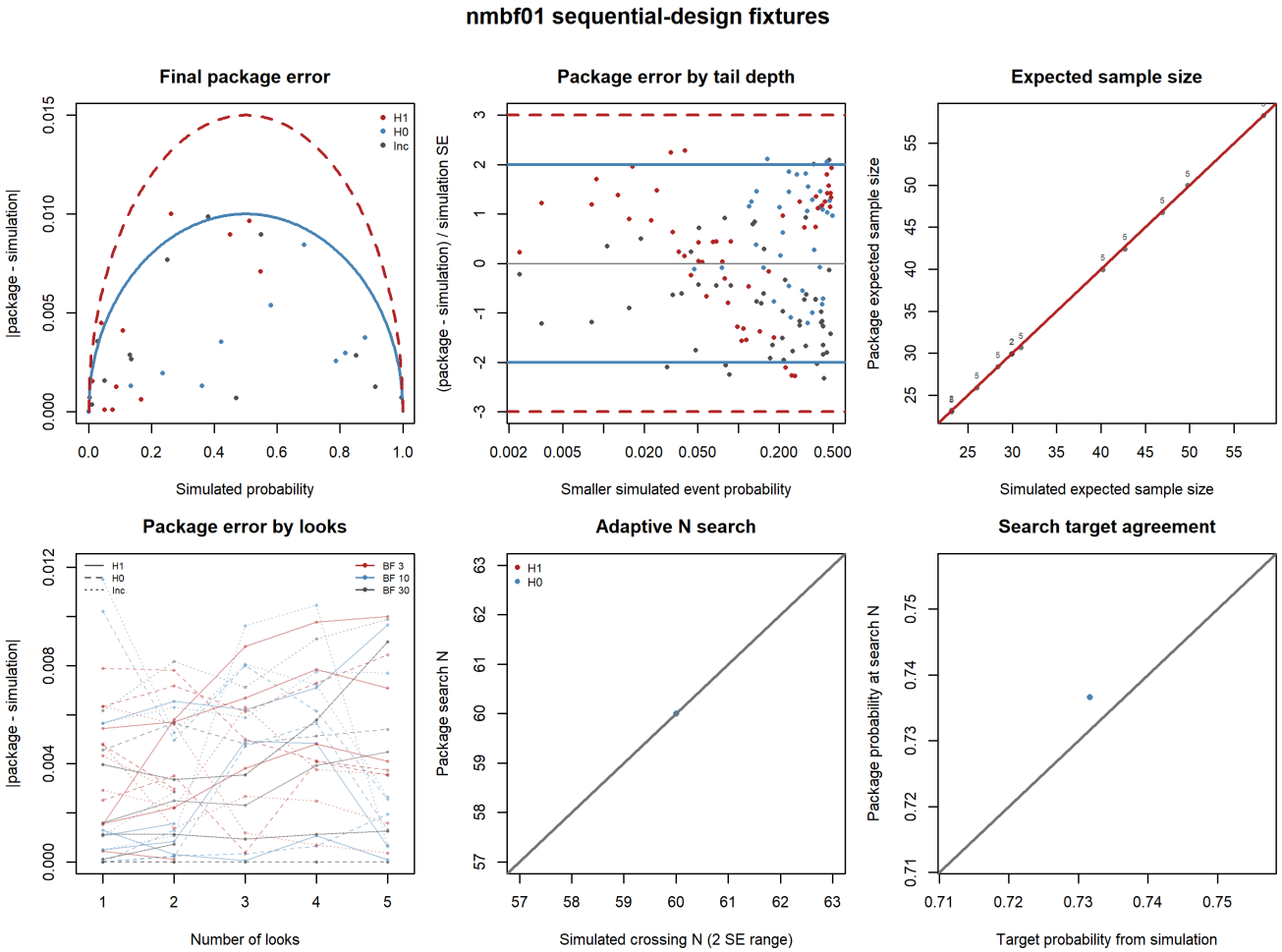
Sequential fixture schedules use looks every 10 observations: one family starts at $n = 20$ and includes 2–20 looks; the stress schedules start at $n = 10$ and include 50, 100, or 200 looks. The 100- and 200-look schedules are compatible with the longer design sequence only.

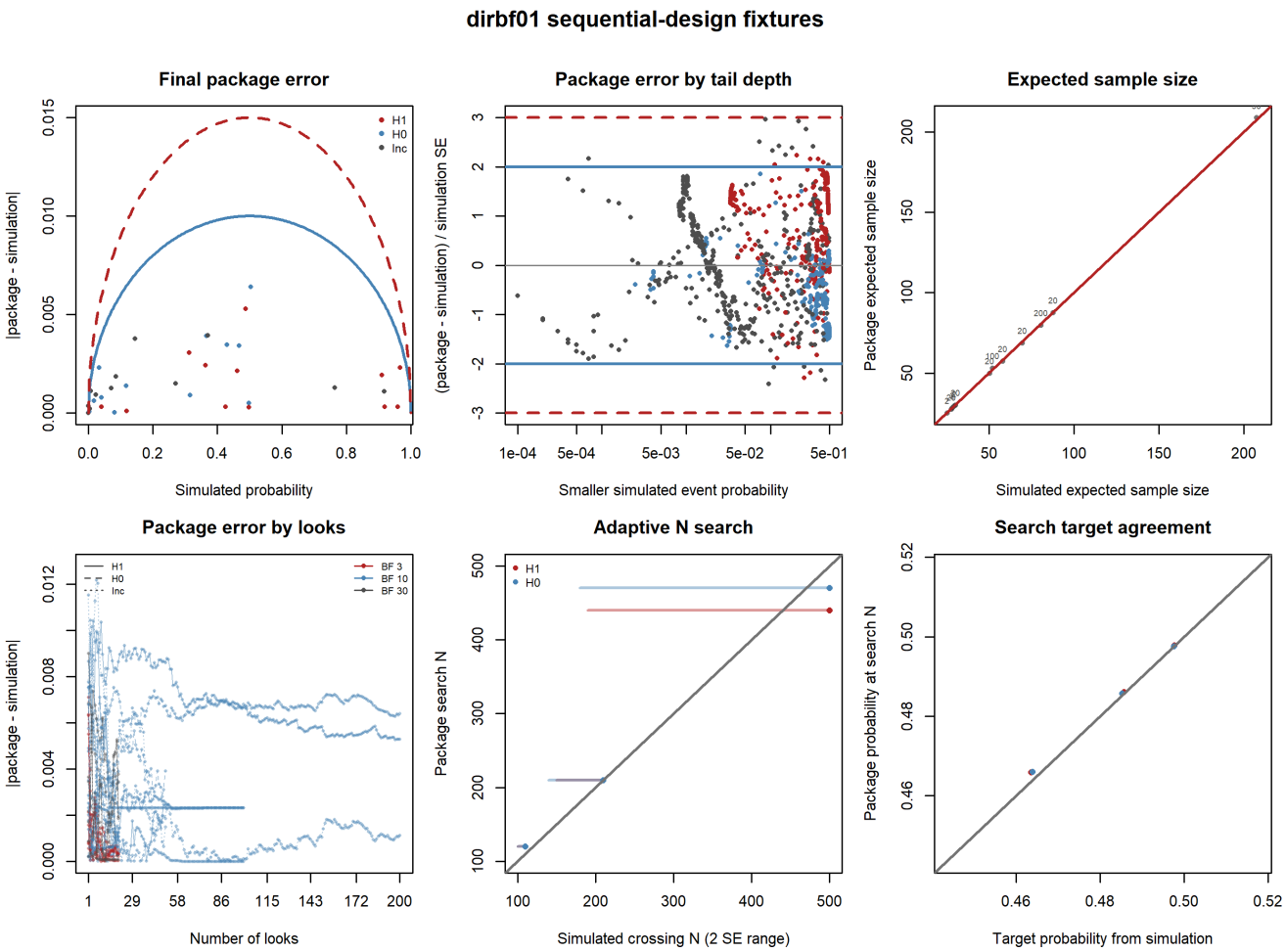
Table 4: Sequential-design fixture setups for z-Test analyses.

Function	Analysis priors	Evidence thresholds	Look sequence
bf01	73	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
nmbf01	35	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
dirbf01	23	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)

Sequential-design plots validate package probability calculations and adaptive sample-size searches against selected simulated fixture summaries. The panels show final package error, package error by tail depth, expected sample size agreement, design-level package-error traces by number of looks, package adaptive-search N against the first simulated crossing N for the target probability with a two-standard-error simulation range, and agreement between the simulated target probability and the package probability at the returned N. Expected sample size points use the same selected fixture rows as the probability checks and are not aggregated. Package calculations are checked at 2, 5, 10, 20, and 50 looks for normal and directional z-test priors. Moment-prior checks currently cover 2 and 5 looks; the remaining sequential schedules are retained in the settings table and automated fixtures.


bf01 sequential-design fixtures
Final package error
|package - simulation|
Simulated probability
H1
H0
Inc
Package error by tail depth
(package - simulation) / simulation SE
Smaller simulated event probability
Expected sample size
Package expected sample size
Simulated expected sample size
Package error by looks
|package - simulation|
Number of looks
BF 3
BF 10
BF 30
Adaptive N search
Package search N
Simulated crossing N (2 SE range)
Search target agreement
Package probability at search N
Target probability from simulation


nmbf01 sequential-design fixtures
Final package error
|package - simulation|
Simulated probability
H1
H0
Inc
Package error by tail depth
(package - simulation) / simulation SE
Smaller simulated event probability
Expected sample size
Package expected sample size
Simulated expected sample size
Package error by looks
|package - simulation|
Number of looks
BF 3
BF 10
BF 30
Adaptive N search
Package search N
Simulated crossing N (2 SE range)
Search target agreement
Package probability at search N
Target probability from simulation



2 *t*-Test

Simulated Design Priors

Table 5: Simulated design priors for the t-Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	Sampling	n2/n1
T1	Point(0)	10 to 500 by 10	10000	two-sample	1
T2	Point(0.5)	10 to 500 by 10	10000	two-sample	1
T3	Normal(0.5, 0.1)	10 to 500 by 10	10000	two-sample	1
T4	Normal(0.5, 0.5)	10 to 500 by 10	10000	two-sample	1
T5	Point(0)	10 to 500 by 10	10000	one-sample	
T6	Point(0.5)	10 to 500 by 10	10000	one-sample	
T7	Point(0.2)	10 to 500 by 10	10000	one-sample	
T8	Normal(0.5, 0.1)	10 to 500 by 10	10000	one-sample	
T9	Normal(0.2, 0.5)	10 to 500 by 10	10000	one-sample	
T10	Point(0)	10 to 500 by 10	10000	paired	
T11	Point(0.5)	10 to 500 by 10	10000	paired	
T12	Point(0.2)	10 to 500 by 10	10000	paired	
T13	Normal(0.5, 0.1)	10 to 500 by 10	10000	paired	
T14	Normal(0.2, 0.5)	10 to 500 by 10	10000	paired	
T15	Point(-0.4)	10 to 500 by 10	10000	two-sample	1

T16	Point(-0.4)	10 to 500 by 10	10000	one-sample	
T17	Point(-0.4)	10 to 500 by 10	10000	paired	
T18	Point(0)	10 to 500 by 10	10000	two-sample	1.1
T19	Point(0.35)	10 to 500 by 10	10000	two-sample	1.1
T20	Normal(0.35, 0.2)	10 to 500 by 10	10000	two-sample	1.1
T21	Normal(0, 0.5)	10 to 500 by 10	10000	two-sample	1.1
T22	Point(-0.35)	10 to 500 by 10	10000	two-sample	1.1
T23	Point(0)	10 to 500 by 10	10000	two-sample	2
T24	Point(0.35)	10 to 500 by 10	10000	two-sample	2
T25	Normal(0, 0.5)	10 to 500 by 10	10000	two-sample	1
T26	Point(0)	10 to 10000 by 10	10000	two-sample	1
T27	Point(0.2)	10 to 10000 by 10	10000	two-sample	1
T28	Point(0.2)	10 to 10000 by 10	10000	one-sample	
T29	Point(0.2)	10 to 10000 by 10	10000	paired	
T30	Point(0.2)	10 to 10000 by 10	10000	two-sample	1.1
T31	Normal(0.2, 0.1)	10 to 10000 by 10	10000	two-sample	1
T32	Normal(0.2, 0.5)	10 to 10000 by 10	10000	one-sample	
T33	Normal(0.2, 0.5)	10 to 10000 by 10	10000	paired	
T34	Normal(0.2, 0.5)	10 to 10000 by 10	10000	two-sample	1
T35	Normal(0.2, 0.5)	10 to 10000 by 10	10000	two-sample	1.1

Analysis Priors

Table 6: Analysis priors used for the t-Test verification run. Designs lists the matching short IDs from the design-prior table.

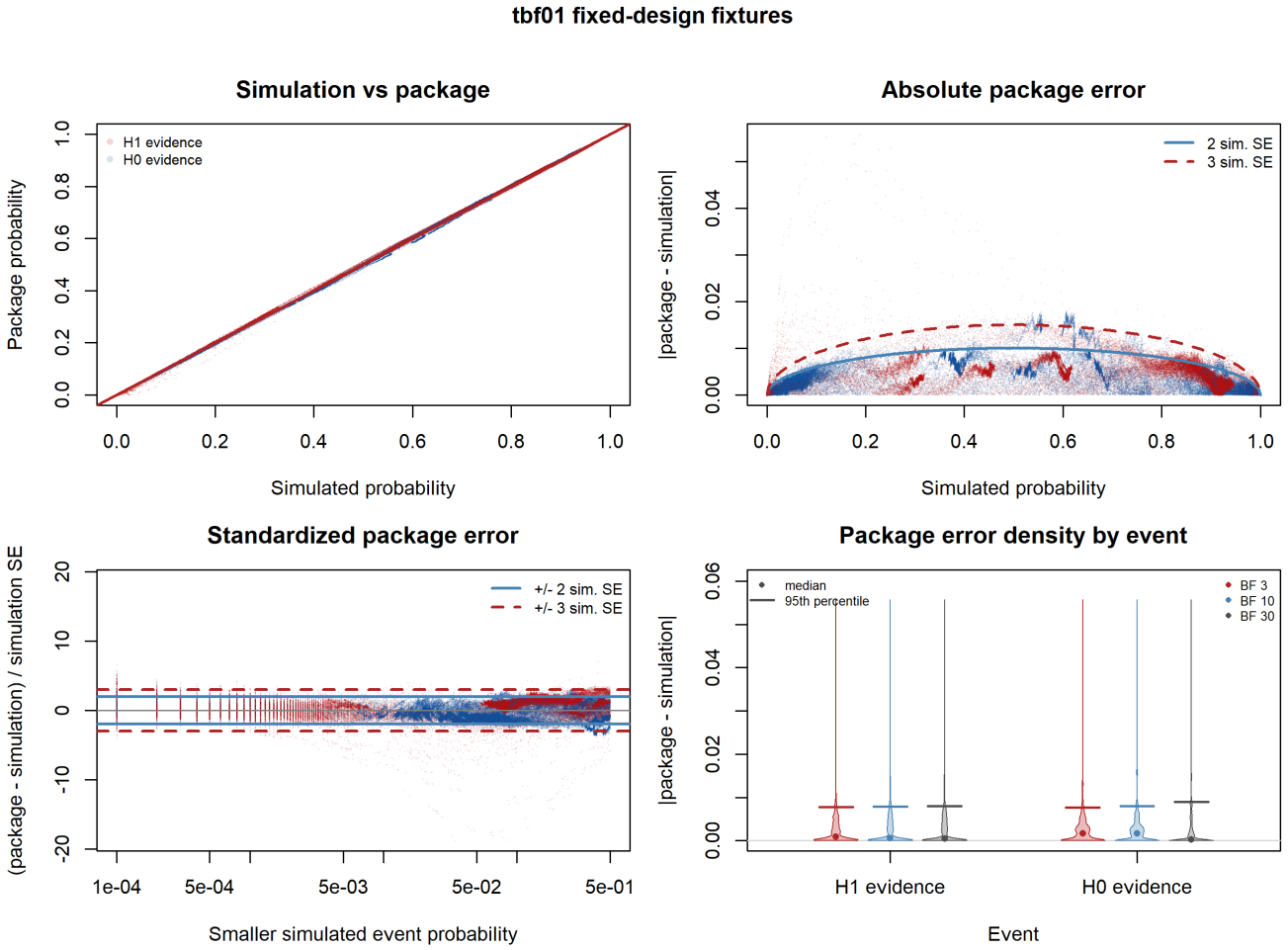
Function	H1	H0	Designs
tbf01	Cauchy$_-$(0, 0.707)	Point(0)	T1-T35
tbf01	Cauchy$_+$(0, 0.707)	Point(0)	T1-T35
tbf01	Cauchy(0, 0.35)	Point(0)	T1-T35
tbf01	Cauchy(0, 0.707)	Point(0)	T1-T35
tbf01	Cauchy(0, 1)	Point(0)	T1-T35
tbf01	StudentT$_+$(3, 0.5, 0.1)	Point(0)	T2-T4, T6-T9, T11-T17, T19-T20, T22, T24, T27-T35
tbf01	StudentT$_+$(3, 0.5, 0.35)	Point(0.2)	T7, T9, T12, T14, T19-T20, T24, T27-T35
tbf01	StudentT$_+$(30, 0.5, 0.35)	Point(0.2)	T1, T5, T10, T18, T23, T26
tbf01	StudentT(3, 0, 0.707)	Point(0)	T1-T35
tbf01	StudentT(30, 0, 0.707)	Point(0)	T1-T35

2.1 Fixed-Design

Table 7: Fixed-design fixture setups for t-Test analyses.

Function	Analysis priors	Evidence thresholds
tbf01	294	BF 3, 10, 30

The fixed-design plots validate package probability calculations against the simulated event frequencies. Cached diagnostic rows are shown; for computationally expensive reference families, the plotted cache can be a stratified subset of the full fixture summary. The panels compare simulated and package probabilities, show absolute and standardized package error, and show package-error densities by event.



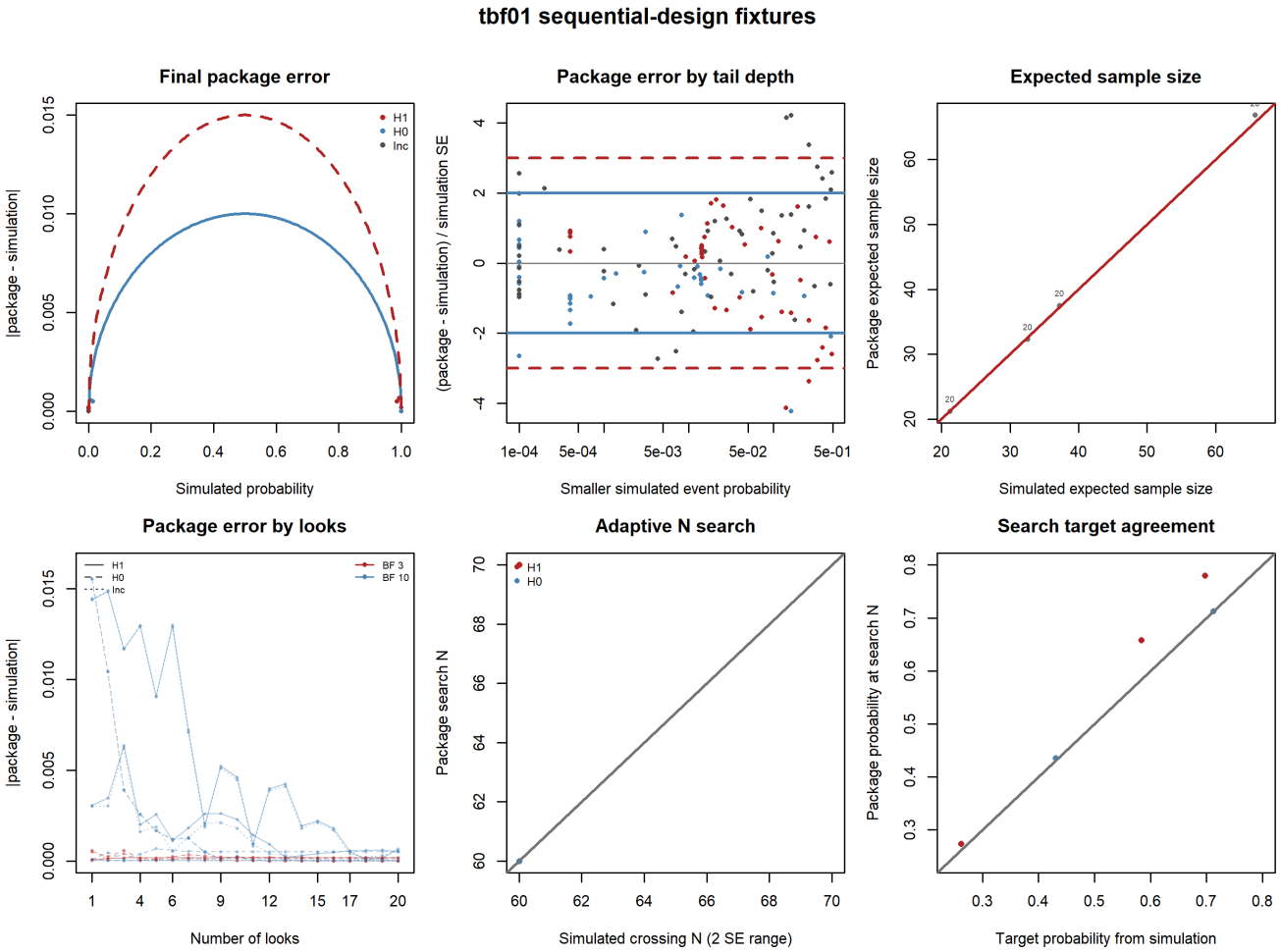
2.2 Sequential-Design

Sequential fixture schedules use looks every 10 observations: one family starts at $n = 20$ and includes 2–20 looks; the stress schedules start at $n = 10$ and include 50, 100, or 200 looks. The 100- and 200-look schedules are compatible with the longer design sequence only.

Table 8: Sequential-design fixture setups for t-Test analyses.

Function	Analysis priors	Evidence thresholds	Look sequence
tbf01	294	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)

Sequential-design plots validate selected t-test fixture summaries against strict `ptbf01seq()` calculations and adaptive `ntbf01seq()` sample-size searches. Probability and expected-sample-size checks use curated strict-reference rows: the original 20-look cases plus five-look wrong-direction and unequal-allocation cases from the expanded registry. The adaptive N-search panels compare package search N to the first simulated crossing N for the target probability with a two-standard-error simulation range. They use representative five-look rows spanning two-sided, one-sided, one-sample, paired, equal and unequal two-sample, shifted-null, wrong-direction, diffuse-design, and long-grid settings; 50/100/200-look stress schedules remain covered by the automated fixture integrity checks.



3 Binomial Test

Simulated Design Priors

Table 9: Simulated design priors for the Binomial Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions
B1	Point(0.5)	10 to 500 by 10	10000
B2	Point(0.6)	10 to 500 by 10	10000
B3	Point(0.2)	10 to 500 by 10	10000
B4	Point(0.75)	10 to 500 by 10	10000
B5	Beta(60, 40)	10 to 500 by 10	10000
B6	Beta(1, 1) on [0.5, 1]	10 to 500 by 10	10000
B7	Beta(1, 1) on [0, 0.5]	10 to 500 by 10	10000
B8	Point(0.5)	10 to 10000 by 10	10000
B9	Point(0.55)	10 to 10000 by 10	10000

Analysis Priors

Table 10: Analysis priors used for the Binomial Test verification run. Designs lists the matching short IDs from the design-prior table.

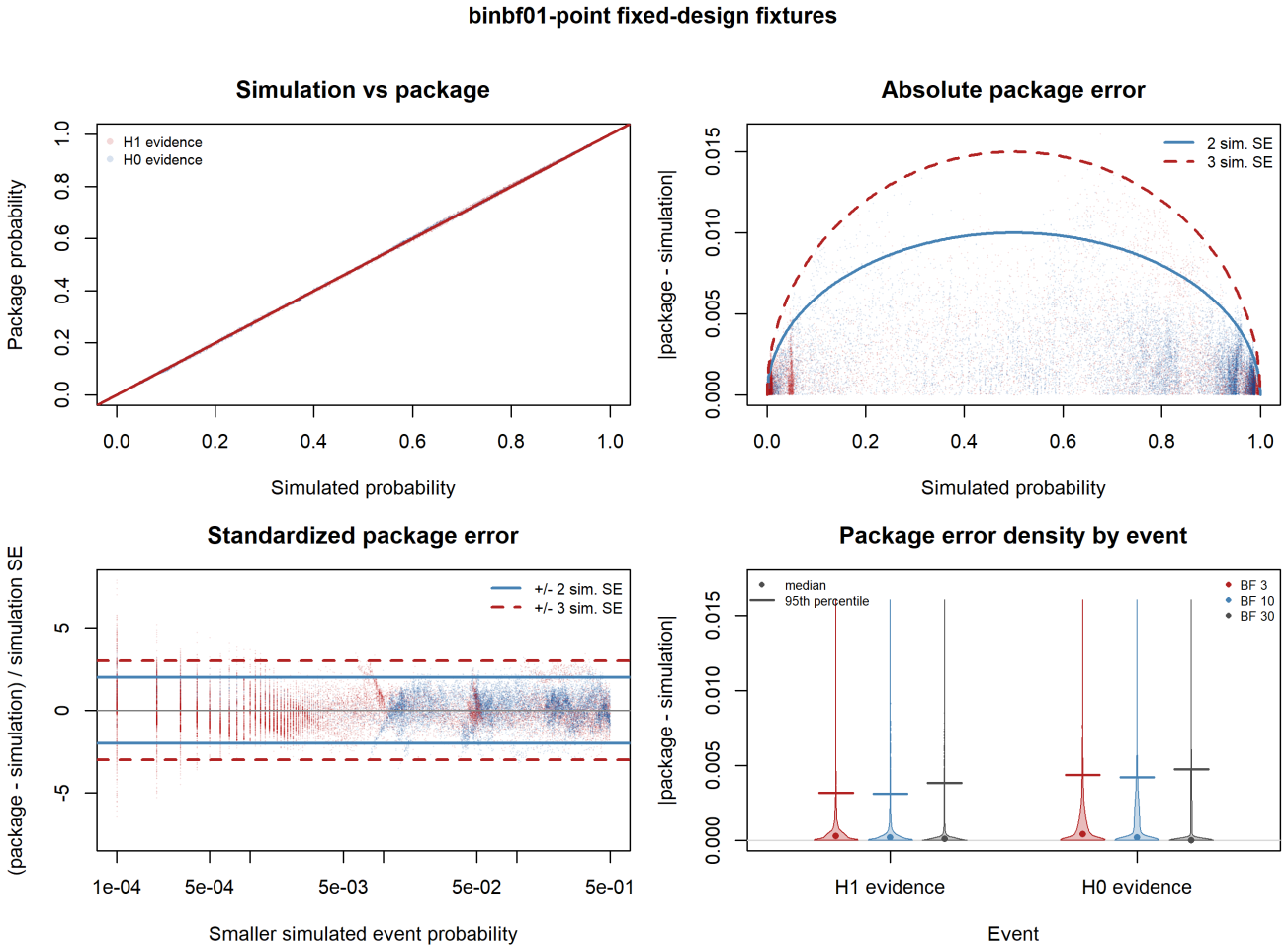
Function	H1	H0	Designs
binbf01 (direction)	$\text{Beta}_+(1, 1)$	Point(0.2)	B3
binbf01 (direction)	$\text{Beta}_+(1, 1)$	Point(0.5)	B1-B9
binbf01 (direction)	$\text{Beta}_+(1, 1)$	Point(0.55)	B9
binbf01 (direction)	$\text{Beta}_+(1, 1)$	Point(0.6)	B2, B5
binbf01 (direction)	$\text{Beta}_+(1, 1)$	Point(0.75)	B4
binbf01 (point)	Beta(1, 1)	Point(0.2)	B3
binbf01 (point)	Beta(1, 1)	Point(0.5)	B1-B9
binbf01 (point)	Beta(1, 1)	Point(0.55)	B9
binbf01 (point)	Beta(1, 1)	Point(0.6)	B2, B5
binbf01 (point)	Beta(1, 1)	Point(0.75)	B4
binbf01 (point)	Beta(40, 60)	Point(0.5)	B1-B9
binbf01 (point)	Beta(5100, 4900)	Point(0.5)	B8-B9
binbf01 (point)	Beta(60, 40)	Point(0.5)	B1-B9

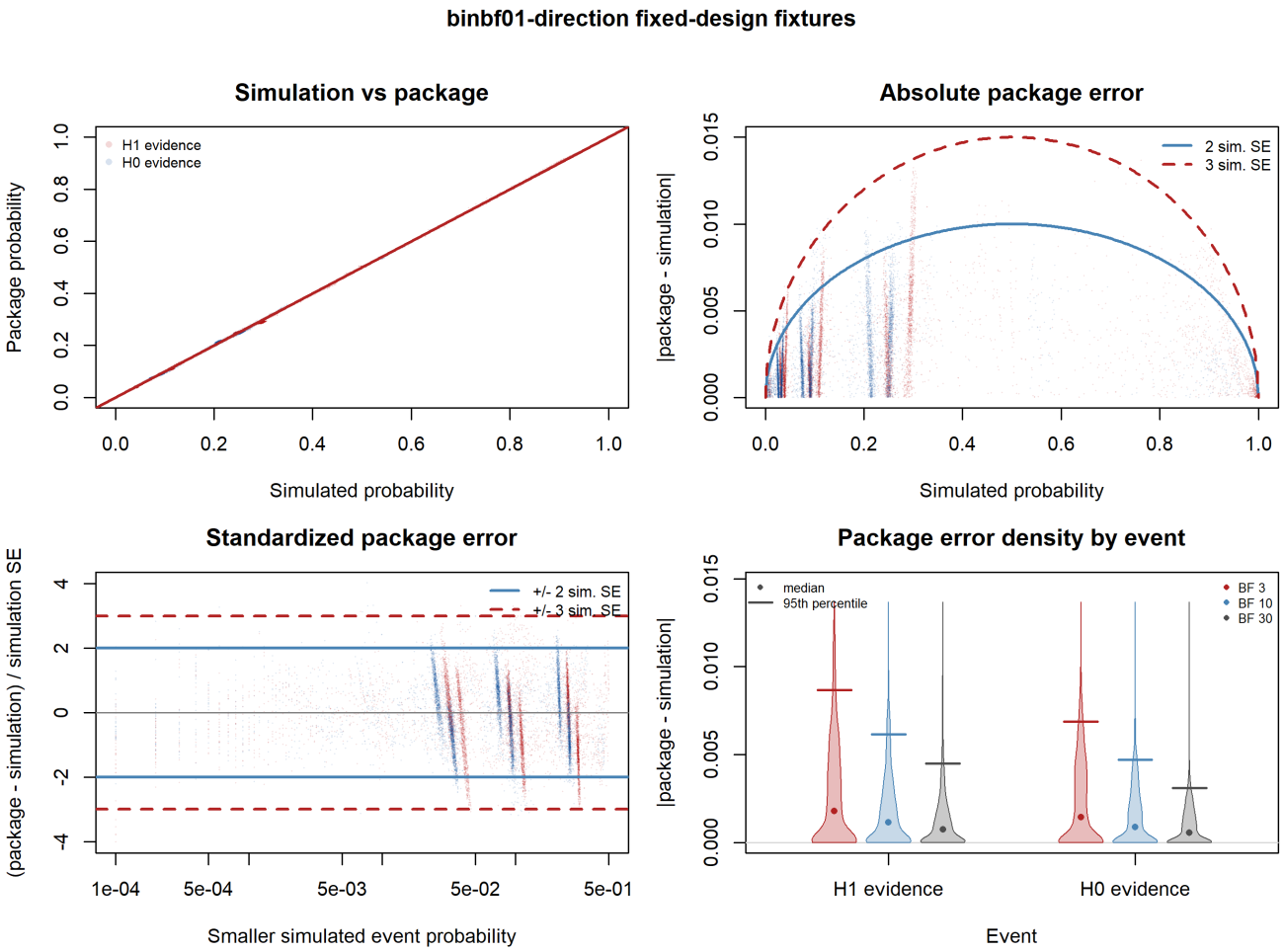
3.1 Fixed-Design

Table 11: Fixed-design fixture setups for Binomial Test analyses.

Function	Analysis priors	Evidence thresholds
binbf01 (point)	34	BF 3, 10, 30
binbf01 (direction)	14	BF 3, 10, 30

The fixed-design plots validate package probability calculations against the simulated event frequencies. Cached diagnostic rows are shown; for computationally expensive reference families, the plotted cache can be a stratified subset of the full fixture summary. The panels compare simulated and package probabilities, show absolute and standardized package error, and show package-error densities by event.


binbf01-point fixed-design fixtures
Simulation vs package
Package probability
Simulated probability
H1 evidence
H0 evidence
Absolute package error
|package - simulation|
2 sim. SE
3 sim. SE
Standardized package error
(package - simulation) / simulation SE
+/- 2 sim. SE
+/- 3 sim. SE
Smaller simulated event probability
Package error density by event
median
95th percentile
BF 3
BF 10
BF 30
Event



3.2 Sequential-Design

Sequential fixture schedules use looks every 10 observations: one family starts at $n = 20$ and includes 2–20 looks; the stress schedules start at $n = 10$ and include 50, 100, or 200 looks. The 100- and 200-look schedules are compatible with the longer design sequence only.

Table 12: Sequential-design fixture setups for Binomial Test analyses.

Function	Analysis priors	Evidence thresholds	Look sequence
binbf01 (point)	34	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
binbf01 (direction)	14	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)

Sequential binomial fixture settings are retained for future package-function validation. The package does not currently expose a sequential binomial evaluation function, so no sequential binomial package-calculation figure is rendered.

4 Validation Summary

The tables and plots in this vignette are a compact summary of the fixture verification suite. They are intended to show which settings were checked, which simulations validate package probability calculations, and which settings are currently retained as integrity-checked fixtures for future function development. The automated package checks remain the authoritative pass/fail record for development.